

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1317

Section 138 kV
June 20, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1317. The Interconnection Customer proposes a 161.8 MW solar photovoltaic facility interconnecting at the Section 138 kV substation in the Virginia area. The request was received on October 25, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 386 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$13,379,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Southwest Substation – Camp Gary 138 kV is overloaded to 175.4% of its 175 MVA emergency rating for the loss of the Steed Road – Tulu 138 kV L. The project contributes 5.1% DFAX (8.2 MW) to this overload.

The Dryden – Camp Gary 138 kV is overloaded to 158.6% of its 183 MVA emergency rating for the loss of the Steed Rd – Tulu 138 kV L. The project contributes 5.1% DFAX (8.2 MW) to this overload.

Donaldson Crk – S Carlyle 138 kV: 148.6% loading (205 MVA) following the outage of the Van Dom – Virginia Hills 138 kV; DFAX = 56.7%.

For the loss of Southwest – Camp Gary 138 kV, the Steed Rd – Tulu 138 kV loads to 135.7% of its 225 MVA rating. Project DFAX: 5.4%.

Clarendon Substation – Arlington 138 kV: 125.0% loading (259 MVA) following the outage of the Hilgen Substation – Carnation Dairy 138 kV; DFAX = 5.1%.

The Van Dom – Virginia Hills Substation 138 kV is overloaded to 117.1% of its 222 MVA emergency rating for the loss of the Donaldson Crk – South Carlyle 138 kV. The project contributes 40.2% DFAX (65.0 MW) to this overload.

The Fosston Municipal – Section 138 kV is overloaded to 113.9% of its 203 MVA emergency rating for the loss of the Van Dom Substation – Virginia Hills 138 kV. The project contributes 61.9% DFAX (100.2 MW) to this overload.

For the loss of Carnation Dairy – Hilgen 138 kV, the Radnor Hts – Clarendon 138 kV loads to 113.8% of its 237 MVA rating. Project DFAX: 5.1%.

The Browns Crossing – Dryden 138 kV Line is overloaded to 113.3% of its 214 MVA emergency rating for the loss of the Steed Rd – Tulu 138 kV. The project contributes 5.1% DFAX (8.2 MW) to this overload.

For the loss of Virginia Hills – Van Dom 138 kV L, the Donaldson Crk – Fosston Municipal 138 kV loads to 112.6% of its 168 MVA rating. Project DFAX: 61.9%.

Belle Haven Substation – Virginia Hills 138 kV: 106.1% loading (196 MVA) following the outage of the South Carlyle – Donaldson Creek 138 kV; DFAX = 40.2%.

The Bonners Ferry Substation – Radnor Hts Substation 138 kV is overloaded to 104.7% of its 228 MVA emergency rating for the loss of the Carnation Dairy – Hilgen 138 kV Line. The project contributes 5.1% DFAX (8.3 MW) to this

overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Southwest – Camp Gary 138 kV	Rebuild line with higher-capacity conductor	\$990,000	100% (\$990,000)
Dryden – Camp Gary 138 kV	Rebuild line with higher-capacity conductor	\$815,000	100% (\$815,000)
S Carlyle – Donaldson Creek 138 kV	Rebuild line with higher-capacity conductor	\$751,000	43% (\$320,000)
Steed Rd – Tulu 138 kV	Rebuild line with higher-capacity conductor	\$1,608,000	100% (\$1,608,000)
Arlington Substation – Clarendon 138 kV	Rebuild line with higher-capacity conductor	\$2,388,000	100% (\$2,388,000)
Van Dom – Virginia Hills 138 kV	Rebuild line with higher-capacity conductor	\$4,316,000	50% (\$2,181,000)
Fosston Municipal – Section 138 kV	Rebuild line with higher-capacity conductor	\$1,604,000	79% (\$1,267,000)
Radnor Hts – Clarendon 138 kV line	Rebuild line with higher-capacity conductor	\$670,000	100% (\$670,000)
Browns Crossing – Dryden 138 kV Line	Rebuild line with higher-capacity conductor	\$1,470,000	100% (\$1,470,000)
Donaldson Crk – Fosston Municipal 138 kV	Rebuild line with higher-capacity conductor	\$1,026,000	79% (\$811,000)
Belle Haven – Virginia Hills Substation 138 kV	Reconductor line	\$1,064,000	50% (\$538,000)
Bonnors Ferry – Radnor Hts 138 kV	Reconductor line	\$321,000	100% (\$321,000)
	Total allocated to project		\$13,379,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.